

Gandharva Appagere

EXPERIMENTAL PHYSICIST • DETECTORS, DATA & AI

PhD candidate in experimental nuclear & hadron physics, Stockholm University

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PROFILE

Experimental physicist (PhD candidate, Stockholm University) working on a large accelerator experiment at GSI/FAIR. Detector work, particle identification, Monte Carlo simulation, statistical analysis, machine-learning classification, and automated analysis pipelines on HPC. Earlier industry experience in automotive radar algorithms and electromagnetic engineering.

Completing my PhD in 2027 and open to industry R&D and data roles. Sectors of interest: Sensing & defence systems • Measurement technology & metrology • Medical technology & radiation • Autonomous systems • Applied R&D • Quantitative analysis

FROM THE LAB TO INDUSTRY

Detectors & sensor systems

Specify, integrate and debug multi-sensor systems; reason from raw sensor behaviour to system performance.

Signal from noise: statistics & uncertainty

Robust inference, detection and estimation with honest uncertainty budgets for decisions that depend on noisy data.

AI/ML & AI-assisted workflows

Apply ML to physical and sensor data with domain understanding, and use modern AI tools to ship analysis software faster.

Measurement, calibration & instrumentation

Design calibration strategies, characterise instruments and keep measurements traceable and reproducible.

Simulation & Monte Carlo

Build digital twins and synthetic data, run sensitivity and feasibility studies before hardware exists.

Large-scale data & HPC

Engineer data pipelines that scale from a notebook to a compute cluster, with reproducible, reviewable code.

EXPERIENCE

PhD Candidate (Doktorand)

Stockholm University • PANDA@HADES, GSI/FAIR

Aug 2021 – present

Stockholm, Sweden & Darmstadt, Germany

Experimental hadron physics on the HADES spectrometer at GSI and simulation studies for the CBM experiment at FAIR.

- Main analysis responsibility in PANDA@HADES for the $\Lambda\Lambda$ channel in proton–proton collisions at $\sqrt{s} = 3.46$ GeV.
- Took detector shifts during the five-week February 2022 proton beam time and a 2025 pion-beam run; worked with drift-chamber online quality assurance.
- Developed a relative time-of-flight particle-identification method that compensates start-detector timing problems.
- Trained multiclass machine-learning classifiers (ROOT TMVA) to separate $\Lambda\Lambda$ signal from background.
- Built an automated analysis pipeline with systematic variations on the GSI batch farm (Slurm / HPC), developed with large language models.
- Wrote an event generator and detector-response folding for CBM feasibility studies of the $\Lambda\Lambda$ spectrum and the ΞN cusps; contributed to the QCD at FAIR white paper.

C++ • ROOT • Python • TMVA • Slurm • PLUTO • GEANT • KinFit • Git

Electric Propulsion Engineer

KTH Hyperloop team

Sep 2018 – 2019

Stockholm, Sweden

High-energy propulsion unit for the SpaceX Hyperloop Pod Competition 2019.

- Helped design a high-energy propulsion unit, a double-sided linear induction motor for the pod; electromagnetic finite-element simulations in COMSOL and ANSYS Maxwell, with MATLAB.

COMSOL • ANSYS Maxwell • MATLAB

Associate Software Engineer

Robert Bosch Engineering and Business Solutions (RBEI)

Aug 2017 – Jul 2018

Bengaluru, India

- Worked on signal-processing algorithms for automotive radar sensors used in driver-assistance systems, in an industrial software-engineering environment.

EARLIER PROJECTS & INTERNSHIPS

Intern, Network Engineering Facility, AT&T, Bengaluru. VPN order-management processes in a network-engineering organisation.

Jan–Jul 2017

Hybrid levitation for Hyperloop: engineering project, BMS College of Engineering, Bengaluru. Design of air-bearing levitation combined with linear-induction-motor propulsion for a Hyperloop concept.

Aug 2016 – Jun 2017

Intern, Quality Assurance division, ISRO, Laboratory for Electro-Optics Systems (LEOS), Bengaluru. Circuit analysis of star, Earth and Sun sensors for the ASTROSAT and Chandrayaan-2 missions.

Jul 2015

EDUCATION

PhD, Experimental Nuclear & Hadron Physics · <i>Stockholm University</i>	2021–2027 (expected)
Licentiate of Philosophy (Fil.lic.), Physics · <i>Stockholm University</i>	2021–2024
Thesis: “ Λ - Λ hyperons: the road to estimate the production cross-section in proton-proton collisions”	
MSc, Electromagnetics, Fusion and Space Engineering · <i>KTH Royal Institute of Technology</i>	2018–2020
BE, Electronics and Communication Engineering · <i>BMS College of Engineering</i>	2013–2017

CORE SKILLS

Data analysis, ML & AI	Statistical inference (maximum-likelihood & template fits, least squares, sideband methods) · Uncertainty quantification (systematic variations, correlated averages) · Kinematic (constrained) fitting (vertex and multi-constraint fits) · Machine-learning classification (multiclass classifiers in ROOT TMVA) · AI-assisted development (large language models for building and automating pipelines)
Programming & software	Python (NumPy, SciPy, pandas, matplotlib, uproot, iminuit, Jupyter) · C++ & ROOT (analysis frameworks, PyROOT, RooFit) · Git / GitLab
Simulation & modelling	Monte Carlo event generation (PLUTO, own generator with final-state interactions) · Detector simulation (GEANT3-based HGeant, CbmRoot) · Finite-element electromagnetics (COMSOL, ANSYS Maxwell)
Detectors & instrumentation	Tracking & timing detectors (drift chambers, straw tubes, RPC, LGAD start detector, time of flight) · Gamma spectroscopy (HPGe detectors, scintillators, neutron activation) · Beam-time operations & online data-quality monitoring · Radiation safety & radioactive-source handling · Automotive radar signal processing
Computing & HPC	HPC batch computing (Slurm on the GSI Virgo cluster) · Containers & software distribution (Singularity/Apptainer, CVMFS) · Large datasets (~155 TB raw data, Lustre, ROOT trees, HDF5)

SELECTED PUBLICATIONS & TALKS

G. Appagere (PANDA@HADES), “Investigating Double-Strange Systems at FAIR: $\Lambda\Lambda$ interactions and neutron star cores”, <i>JPS Conference Proceedings, 15th International Conference on Hypernuclear and Strange Particle Physics (HYP2025), Tokyo</i> . IN PRESS	2026
J. G. Messchendorp, F. Nerling, ..., G. Appagere, et al., “Hadron Physics Opportunities at FAIR”, <i>arXiv:2512.15986 [hep-ex]</i> . PREPRINT	2025
9 conference talks, including NSTAR 2026 (Seville), 15th Nordic Meeting on Nuclear Physics (Visby, 2026), DPG Spring Meeting 2026 (Erlangen) and HYP2025 (Tokyo); plus 6 presentations at internal HADES and CBM collaboration meetings.	
	2022–2026

LABORATORY

Lab responsible, Nuclear physics student laboratory, Fysikum. Handled the radioactive sources for the laboratory through these five years. Responsible for HPGe detectors, scintillators and the source inventory; lab rules and safety, a source booking and record system, and procurement of relevant laboratory components, including detectors and NIM units.	2021–2025
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LEADERSHIP & RESPONSIBILITY

Chair (2025–2026), Vice-Chair (2023–2025), Fysikum PhD Council, Stockholm University.	2023–2026
Vice-Chair, Central Doctoral Student Council (CDR), Stockholm University.	2025
Vice-Chair, Natural Sciences Faculty Doctoral Student Council (NFR), Stockholm University.	2023–2024
Doctoral-student representative, Fysikum Department Board and committees (Research Education, Technical Division, Computer, Work Environment).	2023–2026
University teaching, Stockholm University. Laboratory courses and classroom teaching, including lectures, tutorials and problem solving, in 6 physics courses; co-supervision of a Bachelor’s thesis in Medical Radiation Physics.	2021–2025

AWARDS & LANGUAGES

Awards	SAE E-Baja 2017: 3rd place in India, design review · IEEE Darrel Chong Student Activity Award (2016) · SAE Aero Design West 2015: 13th worldwide (team captain)
Grants	4 grants and scholarships for research travel and experiments (Stockholm University, 2022–2026)
Languages	English: Fluent · Swedish: Basic (A2)